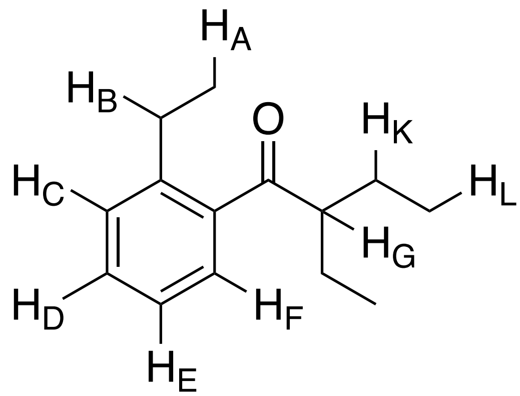
Chem 0345– Organic Lab
Activity 5: NMR Series 2
Extra Practice Problems

1. Consider the molecule below *(Note that not all hydrogens are shown in the structure)*:



   a) Predict the expected splitting pattern of H_A:

   b) Predict the expected splitting pattern of H_B:

   c) Predict the expected splitting pattern of H_C if $J_{CD}>J_{CE}$:

   d) Predict the expected splitting pattern of H_C if $J_{CD}=J_{CE}$:

   e) Predict the expected splitting pattern of H_D if $(J_{CD}=J_{ED})>J_{FD}$

   f) Predict the expected splitting pattern of H_D if $J_{CD}>J_{ED}>J_{FD}$

   g) Predict the expected splitting pattern of H_G:

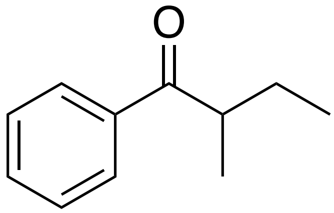
   h) Predict the expected splitting pattern of H_K if $J_{GK}>J_{LK}$

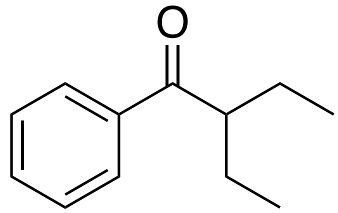
   i) Predict the expected splitting pattern of H_K if $J_{GK}<J_{LK}$

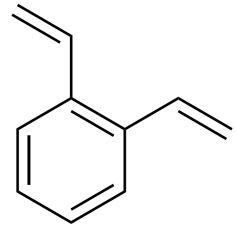
   j) Predict the expected splitting pattern of H_K if $J_{GK}=J_{LK}$

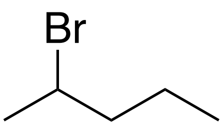
2. How many distinct signals would you expect in the proton NMR of each of the molecules below?

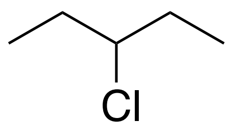
   Note: Assume a high enough resolution that each signal is clearly distinguishable. Also assume that any N-H or O-H protons are observable in the NMR spectrum (if applicable).

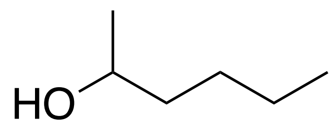


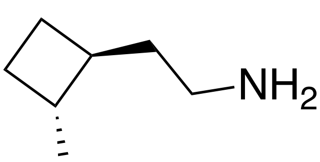


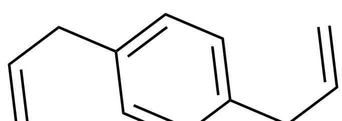


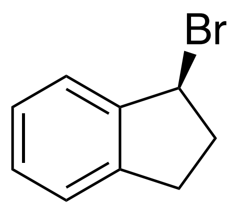


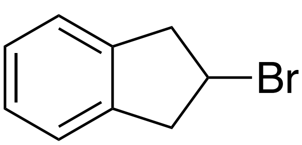


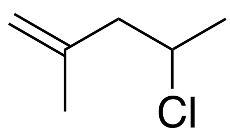


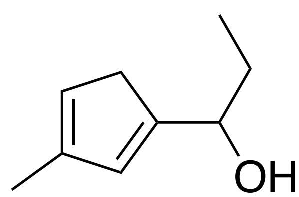




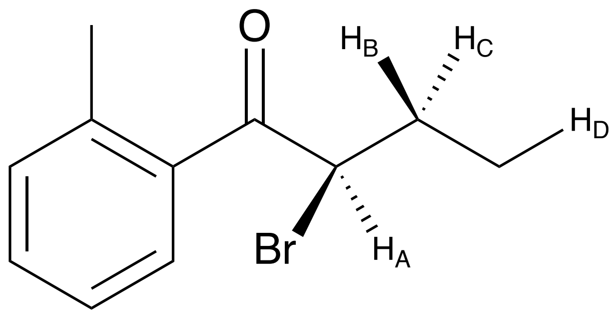








3. Consider the molecule below *(Note: not all hydrogens are being shown. i.e. H_D is just one proton of the methyl group and two of the protons are not being shown)*:



a) Predict the expected splitting pattern of H_B if $J_{CB} > (J_{AB} = J_{DB})$:

b) Predict the expected splitting pattern of H_B if $J_{CB} > J_{AB} > J_{DB}$:

c) Predict the expected splitting pattern of H_B if all coupling constants are equal.

d) Predict the expected splitting pattern of H_A if $J_{BA} > J_{CA}$:

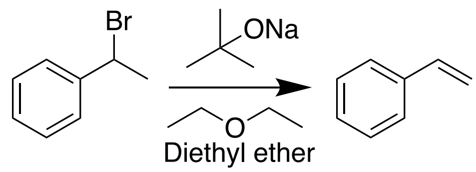
e) Predict the expected splitting pattern of H_A if $J_{BA} = J_{CA}$:

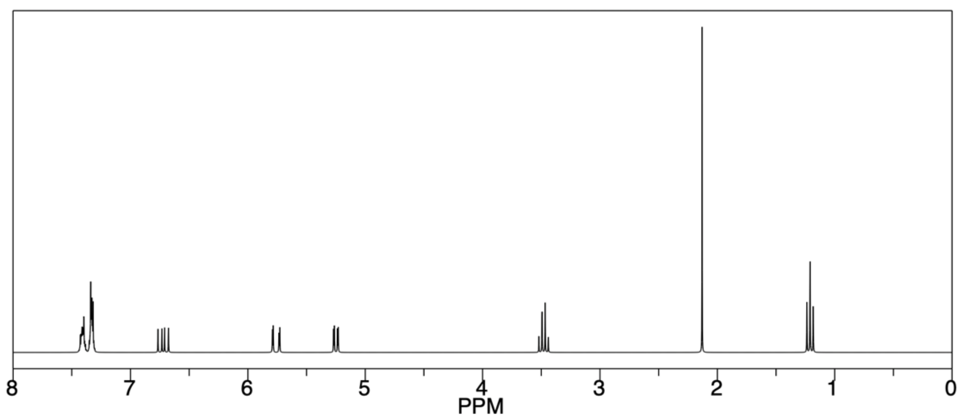
f) Predict the expected splitting pattern of H_D if all coupling constants are equal:

g) Predict the expected splitting pattern of H_B if 2- and 3-bond couplings are observed and all 3-bond coupling constants are equal but $J_{2\text{-bond}} > J_{3\text{-bond}}$:

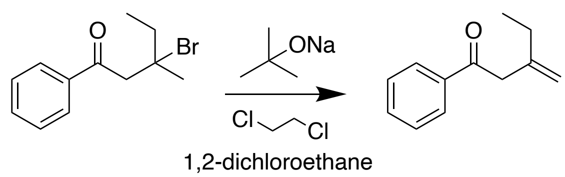
h) If the splitting pattern of H_D is observed to be a doublet of doublets, what can you conclude about the coupling constants?

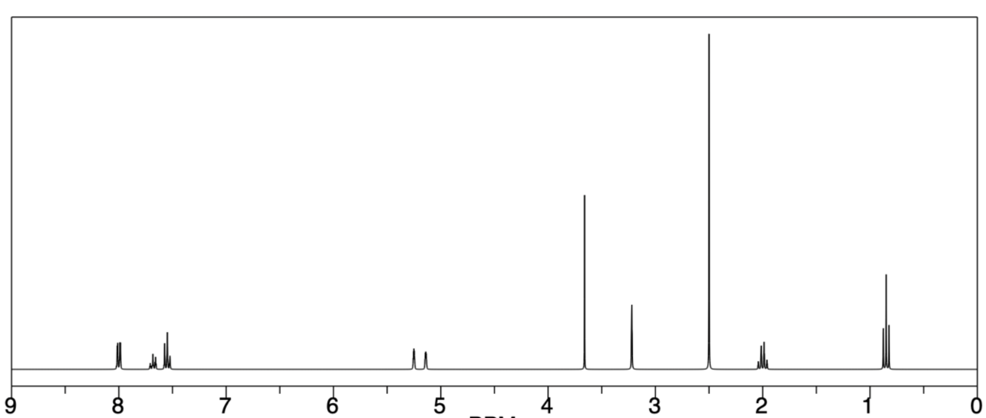
4. At the end of the following elimination reaction, an H-NMR spectrum of the crude product was acquired in $(CD_3)_2CO$. Identify all residual solvent signals from either the reaction solvent or the deuterated solvent used to prepare the NMR sample. *You may assume that all the starting reactants were consumed to form the desired product.*



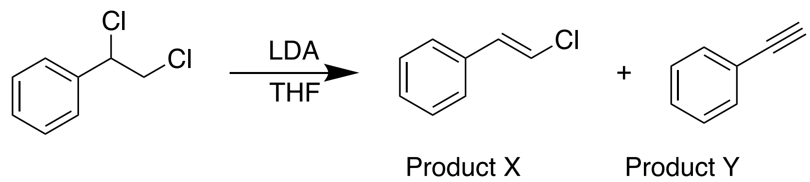


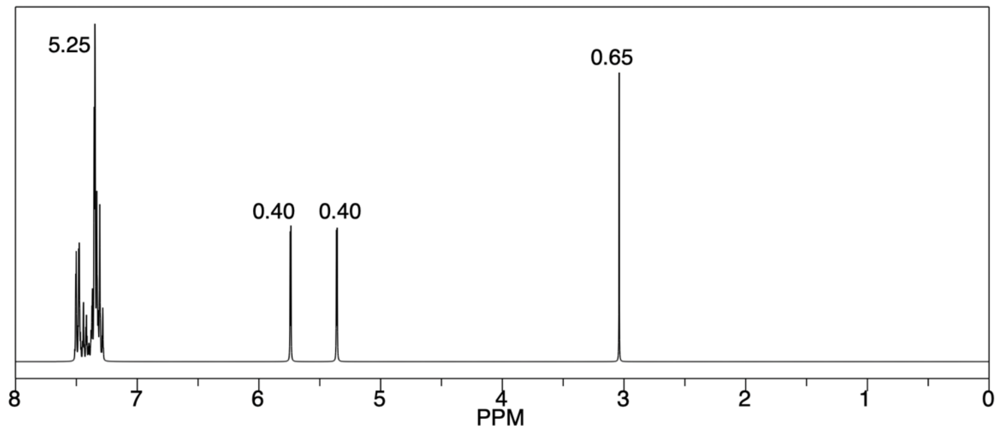
5. At the end of the following elimination reaction, an H-NMR spectrum of the crude product was acquired in $(CD_3)_2SO$. Identify all residual solvent signals from either the reaction solvent or the deuterated solvent used to prepare the NMR sample. *You may assume that all the starting reactants were consumed to form the desired product.*





6. The following reaction resulted in a mixture of products (X and Y). Use the NMR spectrum to calculate the percentage of products X and Y. *You may assume that no starting materials are present in the spectrum. For clarity, the numbers above each signal are the integrations.*





7. Use the NMR spectrum to calculate the percentage of products X and Y. *You may assume that no starting materials are present in the spectrum. For clarity, the numbers above each signal are the integrations.*